

Statistics Spring 2025 HW1

Problem 1: Consider a sample $X_n = \{x_1, \dots, x_n\}$ of i.i.d. observations from **Pareto distribution** with PDF:

$$f_X(x) = \begin{cases} \alpha x_m^\alpha / x^{\alpha+1} & x \geq x_m \\ 0 & \text{otherwise} \end{cases}$$

with (unknown) parameters $x_m > 0$ – called **scale**, and $\alpha > 0$ – called **shape**. Denote $\theta = (\alpha, x_m)$.

1. Find the **method-of-moments estimate** $\hat{\theta}_{\text{MOM}}$. Under what constraint on α does it exist? For further tasks pick α satisfying this constraint, but what happens to $\hat{\theta}_{\text{MOM}}$ when it reaches the boundary of this constraint?
2. Will $\hat{\theta}_{\text{MOM}}$ be **asymptotically normal**? (*Spoiler*: it will, since it's a continuous function of sample moments which are a.n.). Generate a sample X_n in Python (or wherever you like), see the behaviour of $\hat{\theta}_{\text{MOM}}$ as n increases (take, say, $n = 100, 200, 500, 1000, 2000$) – plot Quantile-quantile (QQ) graphs (for that, you need to estimate the variance of the normal distr.-s giving you theoretical quantiles – Python will automatically just estimate μ, σ^2 with sample mean and std to give you a normal distribution providing theoretical quantiles – that's ok).
3. Also find **bootstrap** estimates of the variance of $\hat{\theta}_{\text{MOM}}$ for different sample sizes.
4. Find **Fisher information** matrix. Compare the variance estimate that it provides (actually, rather a lower bound on it – called Cramér-Rao bound) with what you got from bootstrap for different sample sizes.
5. Find the **maximum likelihood estimate** $\hat{\theta}_{\text{MLE}}$. Will $\hat{x}_m$ be asymptotically normal? (It will, but as with the German tank problem, you need $n \gg 1$ to see that). $\hat{\alpha}_{\text{MLE}}$ will be asymptotically normal – compare its error to that of $\hat{\alpha}_{\text{MOM}}$.

Problem 2: Consider a sample $X_n = \{x_1, \dots, x_n\}$ of i.i.d. observations from **Gamma distribution** with PDF:

$$f_X(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}, \quad x > 0$$

with parameters $k > 0$ – called **shape** and $\theta > 0$ – called **scale**.

1. Take the likelihood to exponential form $p(x) = h(x) \exp [\sum_i \eta_i(\theta) T_i(x) - A(\eta)]$. What are the **natural parameters** and the **natural sufficient statistics**? Find the expectation and the variance of x by differentiating $A(\eta)$ – the log partition function.
2. Find the **maximum likelihood estimate** for θ (assume you know k). Using the previous result, find if its **biased** or not.
3. Find **Fisher information** matrix except for the diagonal element corresponding to k . What does its form tell us about k and θ ? Find (the Cramér-Rao bound of) the standard error of your $\hat{\theta}_{\text{MLE}}$ from the previous point.
4. (**Delta method**) Another parameterization of Gamma is with $\alpha = k$ and $\beta = 1/\theta$ – called **rate**. Assume you now want to estimate β rather than θ . Find $\hat{\beta}_{\text{MLE}}$ and its standard error using previous results and the Delta method.
5. * (Extra) Using method of moment estimates (obtained using results of point 1) of k and θ as a starting point, find the MLE estimate of k numerically. (Pick some θ, k , generate a sample, and try ML-Estimating them). Compare the errors of MOM and MLE estimates as sample size increases.